RESP-100 · BASIC EKG INTERPRETATION · WEEK 11

Week 11 Study Guide: Heart Blocks

Walraven, *Basic Arrhythmias* 8th ed. · Chapter 7 (plus Chapter 8 practice) · Lecture by Michael V Hyland (1 h 46 min)

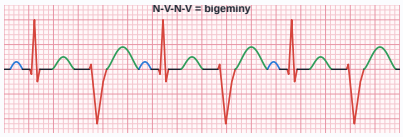
In one line: in every heart block the problem is the **AV node**. It either **holds** the impulse too long (long PRI) or **blocks** it (a P wave with no QRS). **Measure every PRI** and the block will tell you its name.

0 · Ventricular practice recap | 1 · What goes wrong at the AV node | 2 · 1st degree
3 · 2nd degree type I (Wenckebach) | 4 · 2nd degree type II | 5 · 3rd degree (complete)
6 · Side-by-side comparison | 7 · Decision tree | Cheat sheet | Quiz

0 Ventricular rhythm practice recap (0:00-0:56)

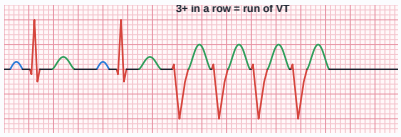
Big idea: the first hour re-taught Chapter 8 live (the screen share wasn't recorded). Full pictures and rules are in the **Week 10 guide**. Here are the reminders and the **grading rules** he added this time.

PVC patterns



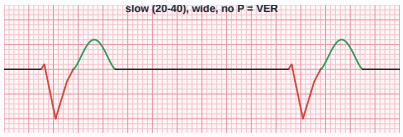
Sinus brady with bigeminy (rate 58). Name the underlying rhythm too.

Run of VT



NSR with a 4-beat run of VT. Rate = underlying rhythm (88).

Ventricular escape



VER / idioventricular: 20-40, wide, no P. Life-threatening.

POINTS YOU WILL LOSE (0:19:28-0:21:17)

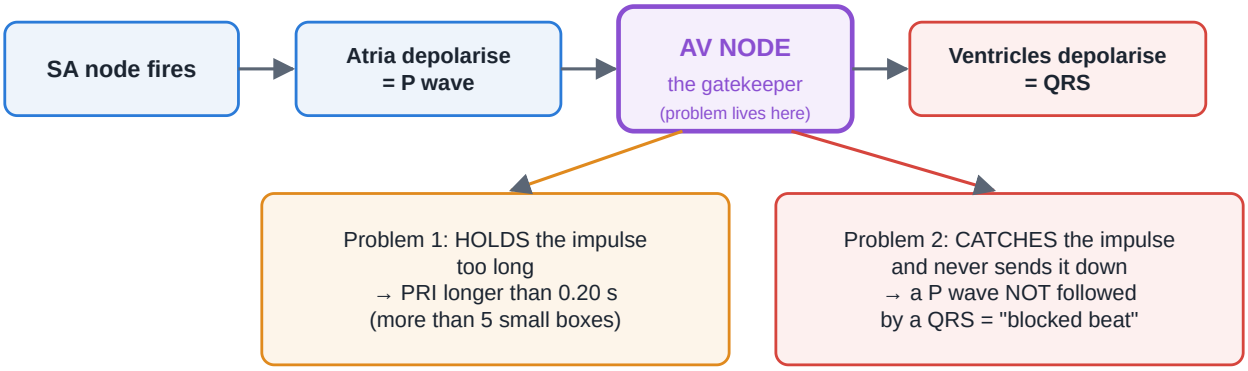
- "NSR with 2 PVCs" without **unifocal / multifocal**: minus 1.
- QRS line without **both** widths (**underlying / PVC**, e.g. 0.08 / 0.16): minus 1. A third number if multifocal PVCs differ in width.
- Regularity: write "**regular with ectopics**" (e.g. "regular with 4 ectopics").

REMEMBER

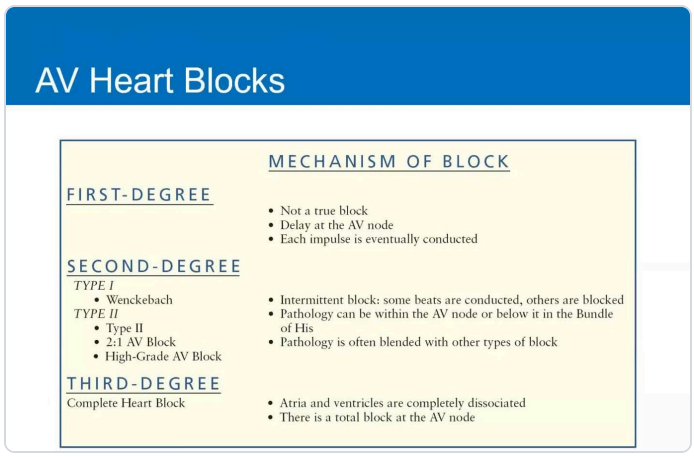
- **Don't count PVCs in the rate**: they may not perfuse.
- VT: 150-250; below 150 = slow VT; above 250 = V-flutter (never tested).
- VF regularity = "**chaotic**". VT & VF = shockable.
- **PEA**: a rhythm on the monitor but no pulse. Call a code.

1 What goes wrong at the AV node

Big idea: heart blocks create only **two** things on the strip: **longer PRIs** and **blocked beats** (a P wave not followed by a QRS). "The PRI is really the key to interpret heart blocks."



The SA node is still the pacemaker. The trouble is the AV node passing the impulse on.



Overview slide: 1st degree = delay only; 2nd degree = intermittent block (type I and type II); 3rd degree = complete block, atria and ventricles dissociated.

Block	Also called
1st degree	Not a true block, just a **delay**
2nd degree type I	**Wenckebach**, Mobitz I (Roman numeral I)
2nd degree type II	**Mobitz II**, "classic" heart block
3rd degree	**Complete heart block (CHB)**, **AV dissociation**

NORMAL PRI REMINDER

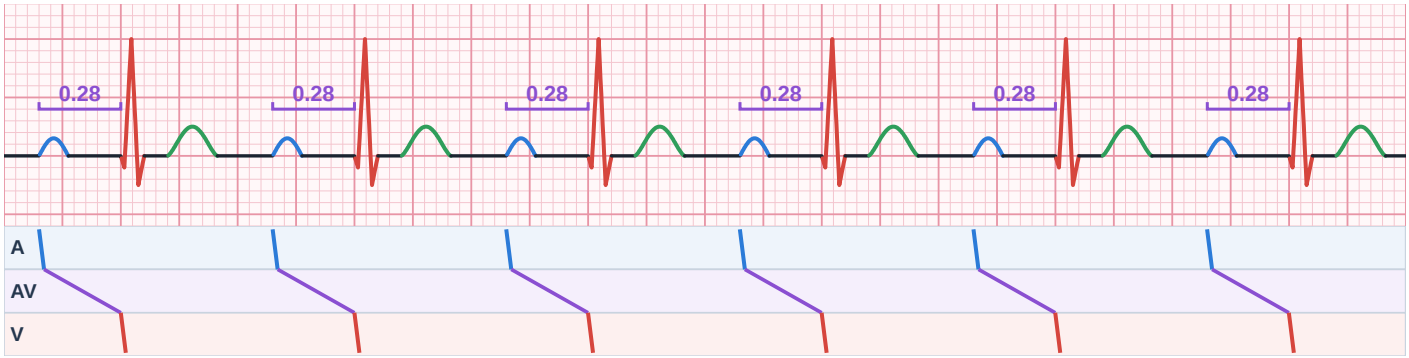
0.12-0.20 s = 3 to 5 small boxes. More than 5 boxes (more than one large box) means the AV node is holding on too long.
Quick check: P starts *before* a large-box line and QRS starts *after* the next one? Then PRI > 0.20.

LADDER DIAGRAMS IN THIS GUIDE

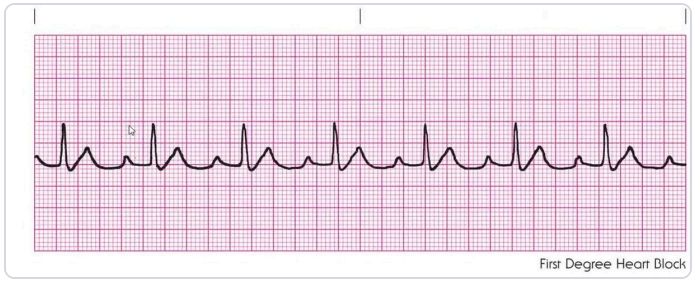
Under each strip: **A** = atria (blue line = P wave), **AV** = the node (purple slope = time spent in the node = PRI), **V** = ventricles (red line = QRS). A red bar across the AV slope = **blocked**. A green dot in V = an **escape** beat starting low down.

2 First-degree heart block

Big idea: the AV node holds **every** impulse too long, but **every impulse gets through**. Every P has a QRS; the PRI is **long (> 0.20) and the same** every beat.



1st degree: every PRI 0.28 (7 boxes), constant. Every P conducted. On the ladder, every AV slope is long but reaches the ventricles.



Lecture strip (1:07:53): R-R 21 boxes, rate 71; PRI 6-6.5 boxes = 0.26; QRS 0.08. **NSR with 1st-degree heart block.**

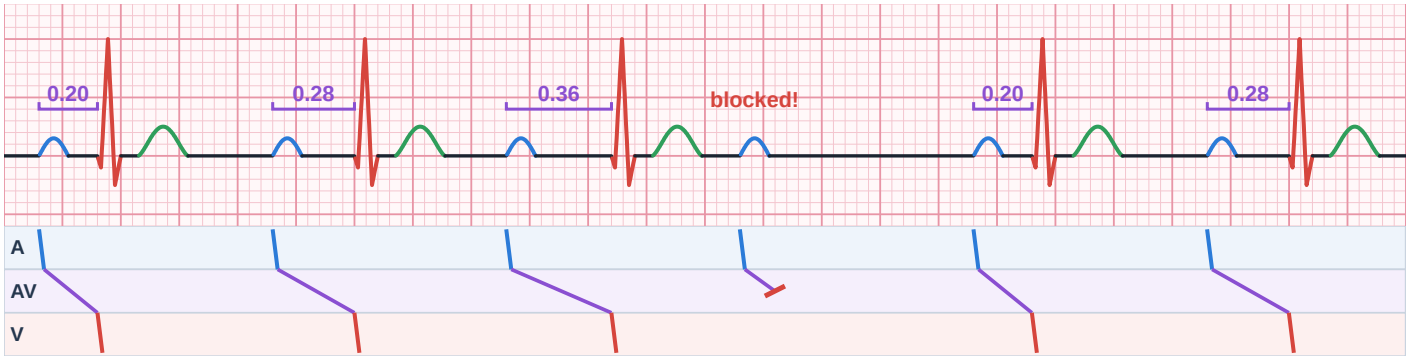
First-Degree Heart Block	
Regularity	Depends on underlying rhythm (usually regular)
Rate	Depends on underlying rhythm
P wave	Upright & uniform; every P followed by a QRS
PRI	**> 0.20 s** (0.21, 0.22 …), **constant**
QRS	< 0.12 s (narrow)

HOW TO NAME IT (1:10:29)

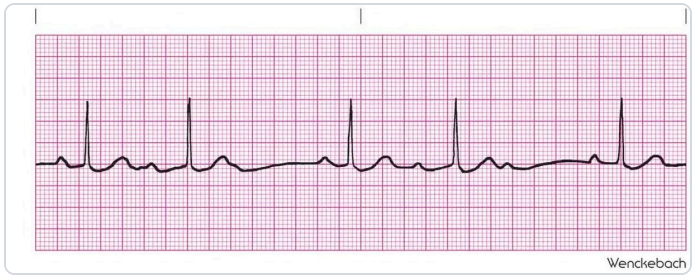
1st degree is the **only** block where you also name the **underlying rhythm**: "normal sinus rhythm with first-degree heart block". Shorthand he accepts: **NSR w/ 1° HB**.

3 Second-degree type I: Wenckebach (Mobitz I)

Big idea: each impulse is held a **little longer than the one before**, so the PRI gets **longer, longer, longer** until one P wave is **blocked**. Then the cycle starts over, "like a broken record".



PRI 0.20 → 0.28 → 0.36 → blocked P → resets to 0.20. On the ladder, the AV slopes get longer until one is cut off.



Lecture strip (1:16:22): groups of 2 beats then a blocked P. Tricky: the blocked P sits on the downslope of a T wave.

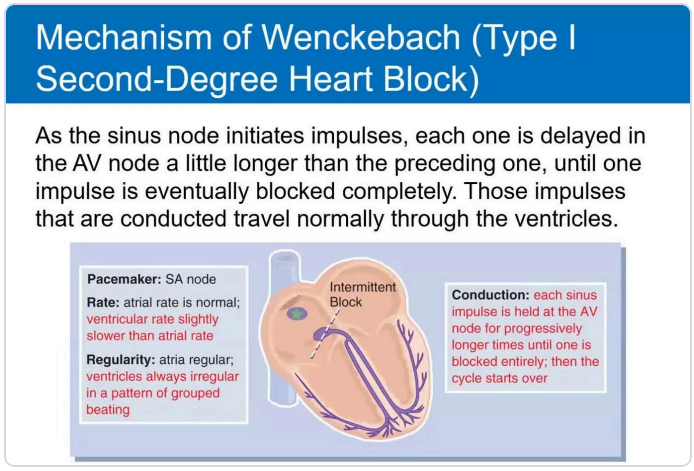
Wenckebach (2nd degree type I)	
Regularity	**Irregular**, "regularly irregular": a pattern of **grouped beating**
Rate	A bit slow. Count R waves in 6 s × 10
P wave	Upright & uniform; some P not followed by QRS
PRI	**Progressively lengthens** until a P is blocked
QRS	< 0.12 s

LECTURE WORKED EXAMPLE (1:17:47)

R-R 23.5, 37, 24.5, 38: irregular. 5 QRS in 6 s gives rate 50. PRIs 6.5 boxes (0.26) then 9 boxes (0.36), then blocked. QRS 0.06.
Write the PRI as a range: shortest-longest, "0.26-0.36".

MEMORY TRICK

"**Longer, longer, longer, DROP: then you have a Wenckebach.**" Why R-R is irregular: a longer PRI pushes each R wave further out, and the dropped beat leaves a gap.



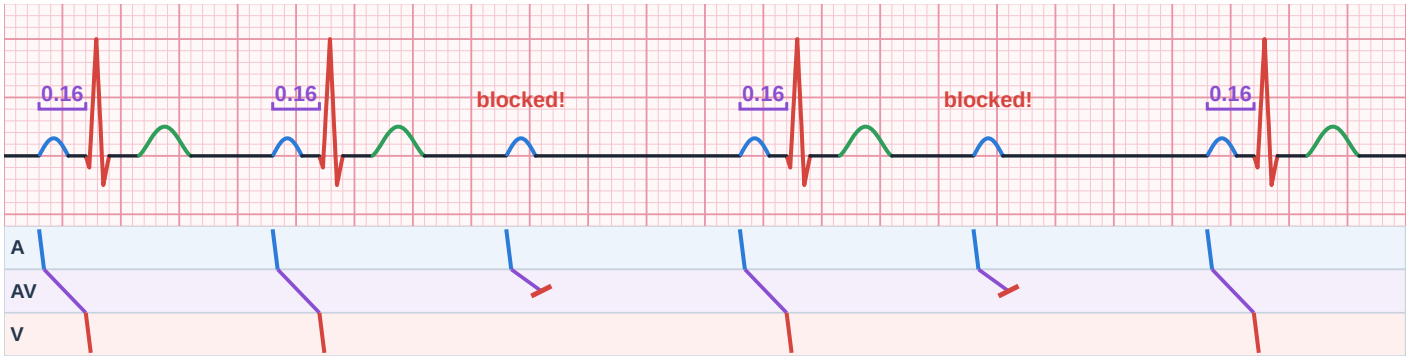
Mechanism: each impulse held progressively longer until one is blocked.

ACCEPTED WORDING

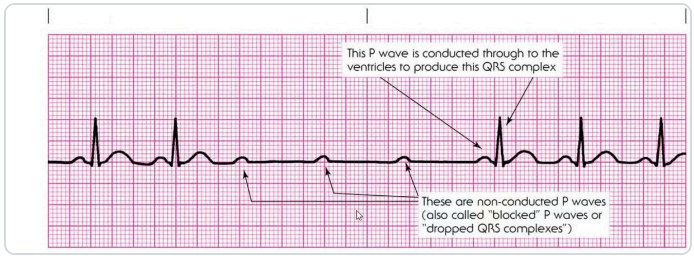
Regularity: "**regularly irregular**" or just "**irregular**". Interpretation: "second-degree heart block type I" or "Wenckebach". **No underlying rhythm needed.**

4 Second-degree type II (Mobitz II, "classic")

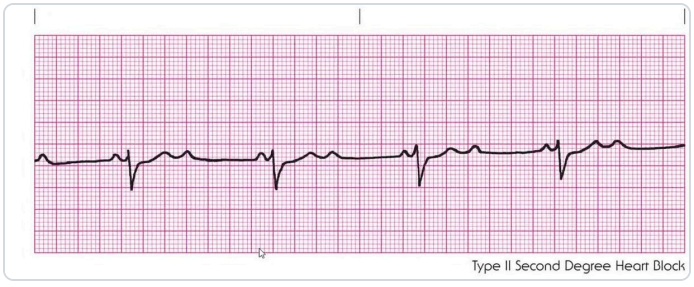
Big idea: the AV node **lets some beats through and blocks others**, without warning. The key: on every **conducted** beat the **PRI stays the same** (normal or long), all the way across the strip. "Blocked beats with a PRI that stays the same: second-degree type II, all day long."



Type II: PRI 0.16 on every conducted beat, then P waves suddenly blocked. No warning lengthening.

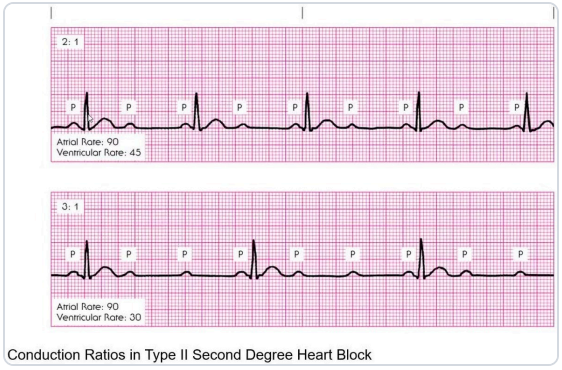


Blocked P waves (1:24:58): 3 non-conducted P waves in a row. PRI 4.5 boxes on every conducted beat.

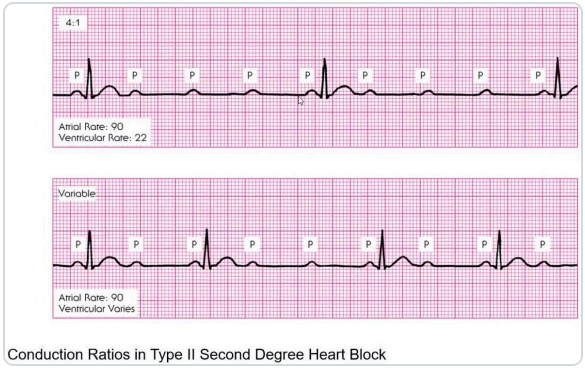


Lecture strip (1:29:10): blocked, conducted, blocked, conducted = **2:1 block**.

Conduction ratios and how to count the rate

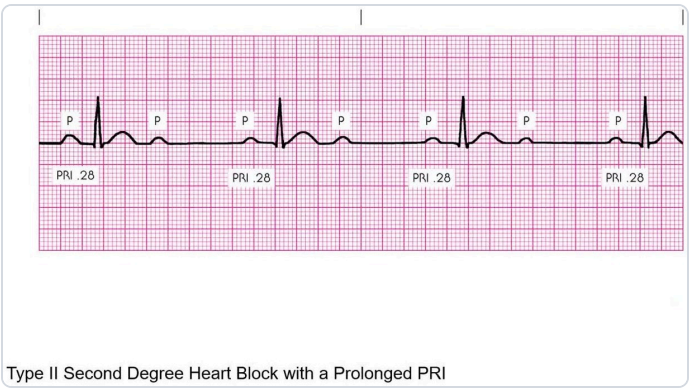


2:1 (atrial 90, ventricular 45) and 3:1 (ventricular 30).



4:1 (ventricular 22) and variable block (irregular).

Block ratio	R-R	Count the rate by	Lecture example
Consistent (2:1, 3:1, 4:1)	Regular	1500 ÷ small boxes between R waves	4:1: 67.5 boxes gives low 20s
Varies (2:1 then 3:1 …)	Irregular	R waves in 6 s × 10	Variable: 4 R waves = 40



Type II with a **prolonged** PRI (0.28), still constant on every conducted beat.

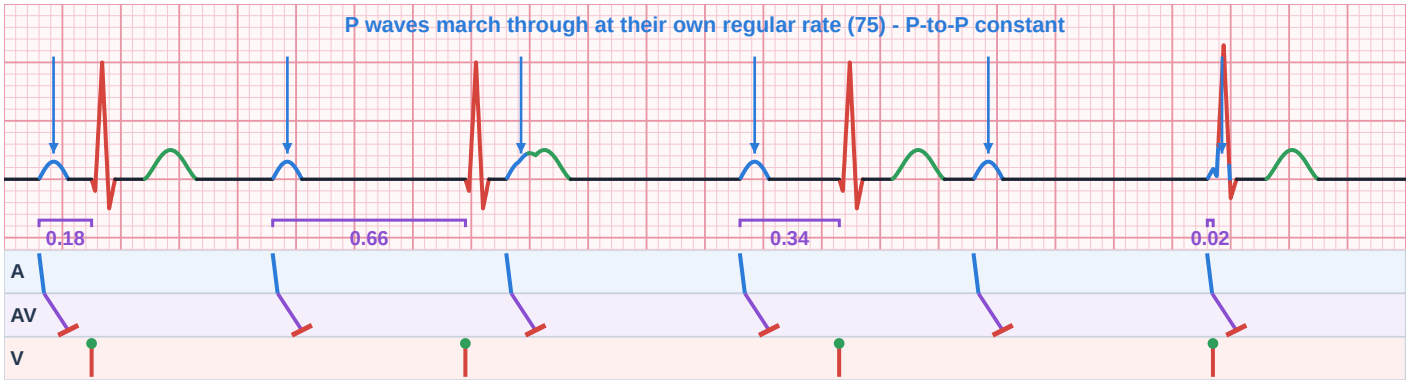
2nd Degree Type II	
Regularity	Regular if ratio consistent; irregular if it varies
Rate	Atrial normal; ventricular slow (½, ⅓, ¼ of atrial)
P wave	Upright & uniform; **more P's than QRS's**
PRI	**Constant** on conducted beats (normal or long) ★
QRS	< 0.12 s

DANGER (1:31:39)

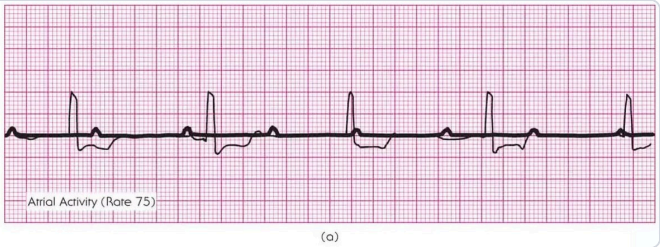
Type II with a ventricular rate **below 60** is life-threatening: call 911 / rapid response. The more beats get blocked, the slower the ventricles and the more dangerous it is.

5 Third-degree / complete heart block (CHB)

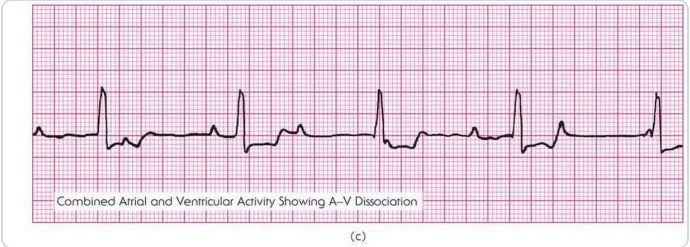
Big idea: the AV node is **completely blocked**: no impulse gets through. The atria keep following the SA node (regular P's), and a **backup pacemaker below the block** (junction or ventricles) paces the ventricles (regular QRS's). The two march along **independently**, so the PRIs are all over the place.



P-P is regular (75) and R-R is regular (47), but they ignore each other. The "PRIs" (0.18, 0.66, 0.34, 0.02) are random, so **no relationship**. P waves hide in T waves and QRSs.



(a) Bold line = atrial activity (rate 75), marching regularly.



(c) Combined strip: lecture example (1:38:15).

LECTURE WORKED EXAMPLE (1:38:24)

R-R 31.5-32 boxes: regular; 1500 ÷ 32 = **47** (the junction's range!). P waves found everywhere: before the QRS, in T waves, on the QRS. P-to-P regular. PRIs 14.5, 6.5, 9, 10, 2 boxes, so "no rhythm has this kind of PRI spread". QRS 2.5 boxes = 0.10, narrow. **Third-degree heart block with a junctional escape rhythm.** For the PRI line write: "**dissociated**" or "**not related**".

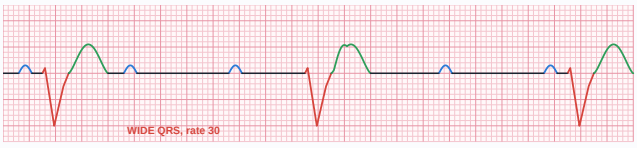
Who is pacing the ventricles? Look at QRS width and rate

CHB with junctional escape



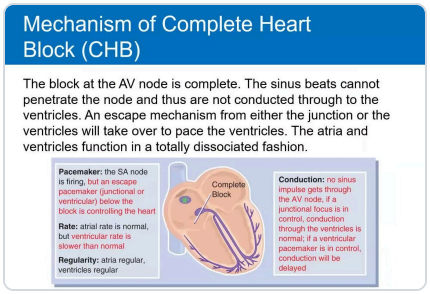
Lower junction fires down the normal pathway, so the QRS is **narrow**; rate **40-60**.

CHB with ventricular escape



Ventricular focus, cell-to-cell, so the QRS is **wide (≥ 0.12)**; rate **20-40**.

Complete Heart Block	
Regularity	**Regular** (P-P regular, R-R regular)
Rate	Atrial normal (60-100); ventricular 40-60 (junctional) or 20-40 (ventricular)
P wave	Upright & uniform; more P's than QRS's
PRI	**No relationship** ("dissociated")
QRS	< 0.12 junctional; ≥ 0.12 ventricular



Mechanism: complete block; escape focus below.

IMMEDIATELY LIFE-THREATENING

Call 911 / code. The escape rate is too slow for an adequate cardiac output.

MARCH OUT THE P'S

Mark two P waves that you can see clearly, then step that same P-P distance across the strip (paper edge or calipers). You will find "missing" P's hiding inside T waves and QRS complexes, landing exactly on schedule.

All four blocks: how to remember them

MEMORY TRICKS

- **1st:** "late but always arrives": long PRI, nothing dropped.
- **Type I:** "longer, longer, longer, drop: Wenckebach".
- **Type II:** "same, same, same, DROP": no warning.
- **3rd:** "P's and Q's don't talk": atria and ventricles divorced.

WHAT THE INSTRUCTOR STRESSED

- "The PRI is really a key to interpret heart blocks." Measure **every** PRI, not just one (1:03:06).
- Blocked beat = **a P wave not followed by a QRS** (1:02:00).
- Heart blocks are "probably the hardest rhythms to interpret". Review within a few days, then again.

6 Side by side: same sinus node, four different AV nodes

Big idea: in all five strips below the SA node fires at exactly the same rate (P every 0.80 s). Only the **AV node** behaves differently. Compare the purple PRI slopes.



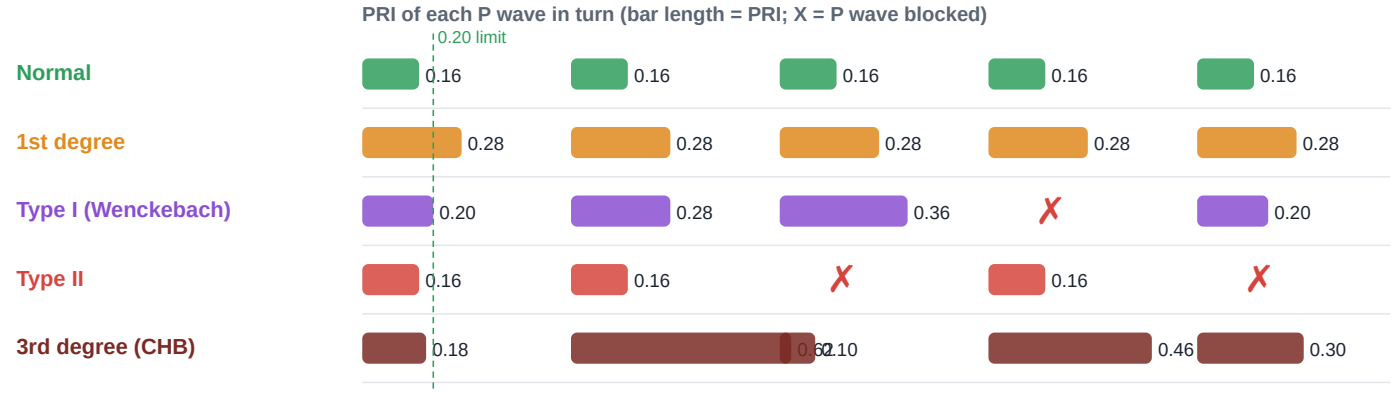
Normal (NSR)
Every P conducted; PRI 0.16, constant.

1st degree
Every P conducted; PRI **long** (0.28) but **constant**.

2nd degree type I (Wenckebach)
PRI **grows** 0.20 → 0.28 → 0.36, then a P is **dropped**. Grouped beating.

2nd degree type II
PRI **constant**; P waves dropped **without warning**.

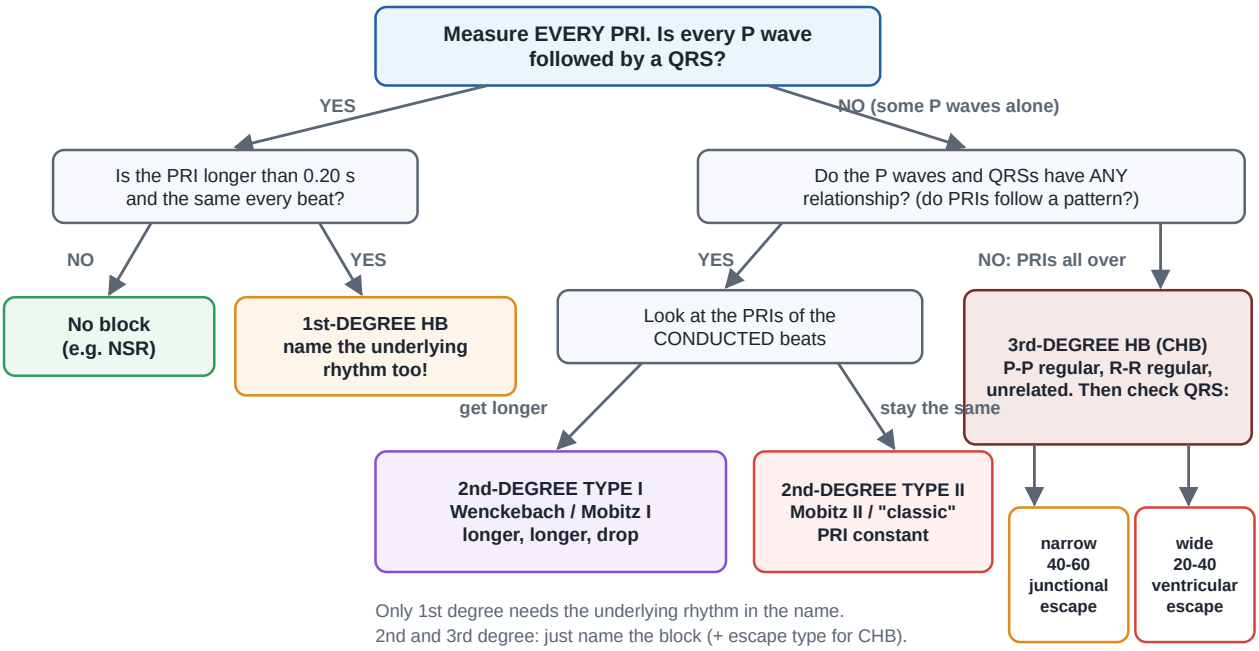
3rd degree (CHB)
Nothing conducted. Ventricles beat on their own (green dot = escape); PRIs random.



The PRI fingerprint of each rhythm: constant-normal, constant-long, growing-then-dropped, constant-with-drops, random.

	1st degree	2nd type I (Wenckebach)	2nd type II	3rd degree (CHB)
Every P followed by QRS?	Yes	No (some dropped)	No (some dropped)	No (none truly conducted)
PRI	> 0.20, constant	Lengthens, then drop	Constant (normal or long)	No relationship
R-R	Regular (per underlying)	Irregular: grouped	Regular if fixed ratio; irregular if varies	Regular
QRS	Narrow	Narrow	Narrow	Narrow (junctional) / wide (ventricular)
Danger	Least	Moderate	Life-threatening if < 60	Immediately life-threatening
Name it as	Underlying rhythm + 1° HB	2nd-degree HB type I	2nd-degree HB type II	3rd-degree HB with junctional / ventricular escape

7 Decision tree: which block is it?



Start at the top. The PRI answers almost every question.

★ Cheat sheet: heart blocks on one page

Rhythm	Regularity	Rate	P wave	PRI (what to write)	QRS
1st-degree HB	Of underlying (usually regular)	Of underlying	Upright & uniform, 1 per QRS	> 0.20, constant (e.g. 0.26)	< 0.12
2nd type I (Wenckebach)	Irregular / regularly irregular	Slightly slow; R's × 10	Upright & uniform; some not followed by QRS	Range, shortest-longest (e.g. 0.26-0.36)	< 0.12
2nd type II	Regular (fixed ratio) / irregular (variable)	Slow; 1500 ÷ boxes if regular, R's × 10 if not	Upright & uniform; more P's than QRS's	Constant on conducted beats (normal or long)	< 0.12
3rd degree (CHB)	Regular	Ventricular 40-60 (junctional) or 20-40 (ventricular)	Upright & uniform; more P's than QRS's	"Dissociated" / not related	< 0.12 junctional; ≥ 0.12 ventricular

The 2 signs of AV-node trouble

1. PRI > 0.20 s (> 5 small boxes) = **delay**
2. P with no QRS = **blocked beat**

Naming rules

1°: underlying + block ("NSR w/ 1° HB")
2°: just "2nd-degree HB type I / type II"
3°: "3rd-degree HB with junctional (or ventricular) escape"

Rate method

R-R regular: **1500 ÷ small boxes**.
R-R irregular (Wenckebach, variable type II): **QRS in 6 s × 10**.
Blocked P's and missing R's don't count; give the **ventricular** rate.

Danger ladder

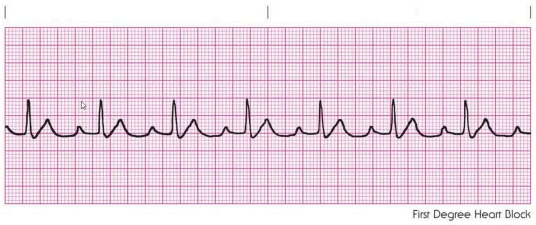
1° (least) < Wenckebach < Type II (**call if < 60**) < CHB (**immediately life-threatening**)

NUMBERS TO OWN

Normal PRI **0.12-0.20** (3-5 boxes) · QRS narrow **< 0.12** · junction **40-60** · ventricles **20-40** · SA **60-100** · 1 small box = 0.04 s · 1 large box = 0.20 s

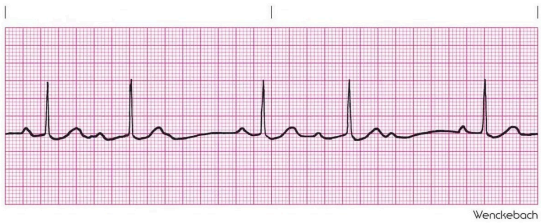
? Self-quiz

1. What two things can a malfunctioning AV node do, and how does each show up on the strip?
2. What is the normal PRI in seconds and in small boxes?
3. Why is first-degree heart block "not a true block"?
4. Which heart block requires you to also name the underlying rhythm?
5. Measure: a PRI covers one large box plus 1.5 small boxes. What is it, and is it normal?
6. In Wenckebach, why is the R-R irregular? What two words describe the regularity?
7. How should you write the PRI for a Wenckebach rhythm whose PRIs are 0.26, 0.32 and 0.36?
8. What single feature separates type I from type II second-degree block?
9. A type II block alternates blocked, conducted, blocked, conducted. What is the ratio, is the rhythm regular, and how do you count the rate?
10. Identify this rhythm (every PRI is 6-6.5 small boxes, R-R 21 boxes).



First Degree Heart Block

11. Identify this rhythm.



Wenckebach

12. In CHB, what do you write on the PRI line, and how do you know the P's and QRS's are unrelated?
13. A CHB strip has a ventricular rate of 47 and QRS 0.10. Name it fully. What if the rate were 30 and the QRS 0.16?
14. Rank the four blocks from least to most dangerous.

Answers

1. (1) Holds the impulse too long, giving a PRI > 0.20 s. (2) Catches the impulse and never sends it down, giving a P wave not followed by a QRS (blocked beat).
2. 0.12-0.20 s = 3-5 small boxes.
3. Every impulse is eventually conducted; it's only delayed, so every P has a QRS.
4. First-degree (e.g. "NSR with 1st-degree HB").
5. 5 + 1.5 = 6.5 boxes × 0.04 = **0.26 s**: prolonged (> 0.20).
6. The lengthening PRI pushes each R further out, and the dropped beat leaves a gap. "Regularly irregular" (or just irregular): grouped beating with a repeating pattern.
7. As a range, shortest-longest: **0.26-0.36**.
8. The PRI of conducted beats: **lengthens** before the drop in type I; **stays constant** in type II.
9. 2:1. The ratio is consistent, so R-R is regular: rate = 1500 ÷ small boxes between R waves.
10. Rate 1500 ÷ 21 = 71, PRI 0.26 constant, every P conducted: **NSR with first-degree heart block**.
11. PRI lengthens, then a P (on the T-wave downslope) is blocked, in groups of 2: **2nd-degree heart block type I (Wenckebach)**.
12. "Dissociated" / "not related". The PRIs vary wildly (e.g. 14.5, 6.5, 9, 10, 2 boxes) while P-P and R-R are each regular.
13. **3rd-degree HB with junctional escape rhythm** (narrow, 40-60). Rate 30 + wide QRS: **3rd-degree HB with ventricular escape rhythm**.
14. 1st degree < type I (Wenckebach) < type II < 3rd degree (CHB).